INCH-POUND
MIL-DTL-24643/46E
22 August 2002
SUPERSEDING
MIL-C-24643/46D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2WAU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG Size 22,7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.024 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification applied by method 3.
Fourth -	Two conductors cabled together with a nominal lay of 1-1/2 inches to form a pair.
Fifth -	The specified number of pairs (see Table I) cabled together with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.
Sixth -	Binder tape applied at manufacturer's option, applied helically with overlap.
Seventh -	Braided shield of tin coated copper.
Eighth -	Binder tape, applied helically with overlap.
Ninth -	Cross-linked polyolefin jacket. Minimum average thickness 0.060 inch.

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TABLE I. Details.

Military part no. M24643/46	Size	Number of pairs	Dia over shielded core (inch nom)	Overall diameter (inches) TYPE LS2WAU min	Overall diameter (inches) TYPE LS2WAU max
-01UO	40	40	1.090	1.320	1.370

EXAMINATION AND TESTS:

Requirements

Basic electrical:
- Conductor resistance - ohms/1000 feet at 25°C, maximum 17.71
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2500
- Insulation resistance - Megohms -1000 feet minimum
  - Conductor to conductor 500
- Jacket flaws ..... No failure
- Conductor and shield continuity ..... No failure

Group A:
- Visual and dimensional ..... No failure
- Capacitance
  - Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum 30
  - Watertightness - See MIL-DTL-24643 for limits of water leakage ..... No failure

Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (when tested at 200°C) 50
- Cold bending, cable - at minus 0 ± 2°C,
  - mandrel diameter 12 times cable diameter ..... No damage
- Drip - 95 ± 1°C ..... Zero
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

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Group C:
Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Shield - conformance to material, construction and coverage ..... No failure
- Surface transfer impedance
  - Milliohms per meter, maximum 700
- Emp response time - dB, minimum 60

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield Insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25

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Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-044)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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